Assignment: Database Administration & Optimization

Scenario: You have been promoted to Junior Database Administrator. You are responsible for keeping the company's database secure, fast, and safe from accidental data loss. Write the SQL commands for the following 4 tasks.

Task 1: Security and Permissions (DCL)

The company just hired a new intern to help with data entry. They need to be able to look up employees and add new ones, but for security reasons, they cannot be allowed to update or delete existing records. **Write the SQL commands to:**

1. Create a new user named `'intern'@'localhost'` with a password of your choice.
2. Grant them **only** the `SELECT` and `INSERT` privileges on the `employees` table.

Task 2: Creating a Virtual Table (Views)

The front desk receptionist needs to look up employee names and their hire dates to organize company anniversaries. However, they should not have access to sensitive information like birth dates or gender. **Write the SQL command to:** Create a `VIEW` named `v_anniversary_list` that selects **only** the `emp_no`, `first_name`, `last_name`, and `hire_date` from the `employees` table.

Task 3: Performance Tuning (Indexes)

The HR web application is running very slowly. You notice that the application constantly searches for employees based on the exact day they were hired, but the database currently has to scan all 300,000 rows every time this search runs. **Write the SQL command to:** Create an `INDEX` named `idx_hire_date` on the `hire_date` column inside the `employees` table to speed up these searches.

Task 4: Safety Nets (TCL Transactions)

You have been asked to test a script that deletes the records of employees who were hired before the year 1986 (`hire_date < '1986-01-01'`). Because this is just a test, you absolutely **must not** permanently delete the data. **Write the SQL script to:**

1. Start a transaction.
2. Run the `DELETE` statement.
3. Immediately undo the deletion to safely restore the database to its original state.